

Soldering Guide – SMD Dice

MakerMotion

Order: **smallest/flattest components first**, THT components (switch, sensor) go last.

Tools

- Fine-tip soldering iron (320–350 °C, lead-free) · 0.5 mm solder wire · flux pen · anti-magnetic tweezers · solder wick · magnifier/microscope · multimeter

Soldering order

#	Component	Note
1	U1 – ATtiny13A-SSU (SOIC-8)	Tin one pad, align IC (watch pin 1!), tack down, solder remaining pins. Check for bridges, ESD care.
2	C1 – 10 µF (0805)	Standard soldering.
3	C5 – 100 nF (1206)	Standard soldering.
4	R2, R3, R4, R9 – 600 Ω (0805)	All four identical, order doesn't matter.
5	LED1–LED7 (1206, yellow)	Watch polarity! Compare cathode stripe with silkscreen. Tin one pad, align, tack, solder second pad.
6	BT1 – CR2032 holder (SMD)	Larger pad, needs a bit more heat/solder.
7	SW2 – tilt switch (THT)	Through-hole part.
8	U3 – SW-460D vibration sensor (THT)	Through-hole part, last.

Test before assembly

- Visual check for solder bridges (IC, closely spaced resistors)
- Multimeter: check continuity between adjacent IC pins
- Insert battery, shake → LEDs should display a number

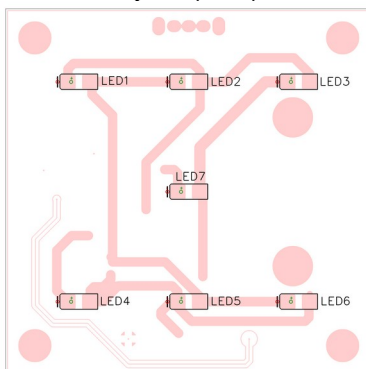
Troubleshooting

Problem	Likely cause
No LED lights up	Check battery polarity, check IC orientation (pin 1)
A single LED doesn't light up	Check that LED's solder joint/polarity
Doesn't react to shaking	Check the SW2/U3 connection to the IC

Assembly diagrams (from Gerber data)

Cathode (–) red, anode (+) green – read directly from the silkscreen layer. U1 pin 1 (green dot) and BT1 polarity are confirmed against the CAD view.

Top side (LEDs)



Bottom side (remaining parts)

